

Knowledge Bank

Acute Myeloid Leukemia Diagnostics — Computational Biology

DAY 1 OF 7

The Problem, The Biology, The Data, The Tools

This document is a learning resource, not a cheat sheet. Every concept is explained from first principles so that you can discuss it naturally, answer follow-up questions, and reason about problems you have not seen before.

For: ML / CS Engineers preparing for ACTREC Project Research Scientist-II interview

Chapter	Topic	What You Will Be Able to Do
1	The Problem — What Is AML?	Explain why this disease is a classification problem
2	The Human Genome — Your Raw Data	Describe DNA, scale, and what you are working with
3	DNA Methylation — Your Core Feature	Explain epigenetics, CpG sites, and why they matter
4	Nanopore Sequencing — Your Data Source	Explain how data is generated and what arrives to you
5	The Computational Pipeline	Trace data from blood sample to model input
6	The ML Task — Framed Properly	Articulate the classification problem with full context
7	Glossary of Terms	Use every term naturally in conversation
8	7 Questions With Deep Answers	Reason through any question, not just rehearse answers

The Problem — What Is Acute Myeloid Leukemia?

Before you can build anything useful, you need to understand what you are building it for. This chapter explains the disease at a level that lets you have an informed conversation with clinicians and justify your engineering choices.

1.1 Blood — A Brief Primer

Blood is not a simple liquid. It is a suspension of specialised cells manufactured continuously by the bone marrow. These cells have distinct jobs and finite lifespans, so the bone marrow must produce billions of new cells every day. This production process is called haematopoiesis.

At the top of the production hierarchy sit haematopoietic stem cells (HSCs) — a small population of multipotent cells that can divide and differentiate into any blood cell type. From HSCs, two main lineages branch off:

Myeloid lineage — gives rise to red blood cells, platelets, neutrophils, monocytes, and eosinophils

Lymphoid lineage — gives rise to T cells, B cells, and natural killer cells

1.2 What Goes Wrong in AML

Acute Myeloid Leukemia is a cancer of the myeloid lineage. The word 'acute' means it progresses rapidly — days to weeks rather than months to years. The word 'myeloid' tells you which lineage is affected. 'Leukemia' literally means white blood in Latin, referring to the white blood cells.

What physically happens is this: a myeloid progenitor cell (an immature cell partway through its development into a mature blood cell) acquires genetic and epigenetic changes that cause it to divide uncontrollably without ever maturing. These abnormal immature cells are called **blasts**. In a healthy person, blasts make up less than 5% of bone marrow cells. In AML, blasts accumulate to more than 20% — sometimes filling the entire marrow.

As blasts crowd the bone marrow, they physically displace the normal cell production lines. The result: the patient runs out of healthy red blood cells (causing severe anaemia and fatigue), healthy platelets (causing dangerous bleeding), and functional immune cells (causing susceptibility to infections). Without treatment, this cascade is fatal within weeks.

Why This Matters for Your Work: AML is not one disease. It is a family of diseases that all look similar under a microscope but have different molecular causes, different responses to treatment, and different survival outcomes. The whole point of this project is to build a system that can identify WHICH specific form of AML a patient has — not just that they have it.

1.3 AML Subtypes — The Classification Problem

The World Health Organisation (WHO) classifies AML into subtypes based on the specific genetic and molecular changes found in the cancer cells. This matters enormously for treatment — some subtypes respond beautifully to standard chemotherapy while others require completely different drugs or stem cell transplants.

The main categories of molecular changes that define subtypes are:

Type of Change	What It Means	Clinical Importance
Chromosomal translocations	Two chromosomes swap segments, creating abnormal fusion genes. Example: t(8;21) joins AML1 and ETO genes	AML with t(8;21) has relatively good prognosis and responds well to standard chemotherapy
Point mutations	A single DNA letter changes in a gene. Example: FLT3-ITD is an insertion in the FLT3 receptor gene	FLT3-ITD = poor prognosis; but targetable with specific FLT3 inhibitor drugs
Gene deletions	A segment of a chromosome is lost. Example: -7 (loss of chromosome 7)	Monosomy 7 associated with therapy-related AML; very poor prognosis
Epigenetic changes	The DNA sequence is unchanged but methylation patterns are abnormally altered	IDH1/IDH2 mutations cause methylation changes; targetable with specific drugs

1.4 Why Diagnosis Is So Difficult Today

Current clinical diagnosis of AML requires a collection of tests run in parallel, each capturing a different layer of information:

Morphology: A pathologist looks at bone marrow cells under a microscope and counts blasts. This identifies AML but cannot distinguish subtypes.

Flow cytometry: Identifies proteins on the cell surface. Helps classify the lineage but misses molecular subtypes.

Cytogenetics (karyotyping): Cultures cells and photographs chromosomes to look for structural changes like translocations. Takes 7-14 days and can fail if cells don't grow.

FISH (Fluorescence In Situ Hybridisation): Tests for specific known chromosomal changes. Fast but only looks for what you already suspect.

Molecular tests (PCR, sequencing): Detects specific gene mutations. Each test is ordered separately, creating delays.

The Vision Behind This Project: Replace this fragmented, multi-step, multi-day process with a single nanopore sequencing run that produces DNA sequence, methylation data, and structural variant information simultaneously — and then run a trained ML classifier that outputs a subtype diagnosis within hours instead of days.

The Human Genome — Understanding Your Raw Data

2.1 What DNA Actually Is

Deoxyribonucleic acid (DNA) is a long polymer — a chain of repeating units called nucleotides. Each nucleotide contains one of four bases: Adenine (A), Thymine (T), Guanine (G), or Cytosine (C). DNA exists as a double helix — two complementary strands wound around each other, held together by base pairing rules: A always pairs with T, and G always pairs with C.

The sequence of these four letters along one strand encodes all the information a cell needs to function. The human genome contains approximately 3.2 billion base pairs. Written out as plain text, the human genome would fill roughly 750 standard novels.

For Your Pipeline: A human genome sequencing run produces a file where the data is literally a string of A, T, G, C characters — roughly 3.2 billion characters per strand, two strands, split across millions of short fragments called reads. Your first job is to reassemble and interpret this string.

2.2 Genes, Exons, Introns, and Regulatory Regions

A gene is a specific region of DNA that encodes the instructions for building a protein. The human genome contains approximately 20,000-25,000 protein-coding genes. However, these genes account for only about 1.5% of the total genome. What is the rest?

Introns: Non-coding sequences within genes that are transcribed but then spliced out before the final protein is made

Regulatory regions: Sequences that control when, where, and how much a gene is expressed. Promoters (just upstream of genes) and enhancers (can be far away) are the most important.

Repetitive elements: Sequences that repeat thousands of times throughout the genome. About 50% of the human genome is repetitive.

Non-coding RNAs: Regions that produce RNA molecules with regulatory functions but no protein product.

Why This Matters for ML: When you extract methylation features from CpG sites, you need to know WHERE those CpGs are. A CpG in a gene's promoter region that gets methylated will silence that gene. A CpG in an intron likely has a different effect. Genomic context is a feature in itself.

2.3 How Genes Become Proteins — The Central Dogma

The flow of biological information follows a principle called the Central Dogma of Molecular Biology: DNA is transcribed into RNA, which is then translated into protein. Understanding this helps you understand what goes wrong in AML and why your measurements matter.

Step	Process	What Happens	Measurement Used In Project
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1	Transcription	A segment of DNA is copied into messenger RNA (mRNA) by RNA polymerase	RNA-seq / Whole Genome Transcriptome Sequencing captures this output
2	RNA Processing	Introns are spliced out; exons joined together; final mRNA produced	Alternative splicing can be a cancer-specific event; detectable in transcriptome
3	Translation	Ribosome reads mRNA and assembles amino acid chain (the protein)	Not directly measured; inferred from RNA levels
4	Epigenetic Regulation	Methylation of promoter DNA determines whether step 1 can even occur	Nanopore methylation calling is your primary data source for this project

2.4 Scale and What It Means Computationally

Quantity	Number	Computational Implication
Base pairs in human genome	~3.2 billion	Raw FASTQ per sample: ~50-150 GB
Protein-coding genes	~20,000-25,000	Upper bound for gene expression features
CpG sites (methylation targets)	~28 million	Dimensionality of raw methylation vector
CpG islands (dense CpG clusters)	~29,000	Biologically important subset of features
Sequencing reads per 30x WGS run	~1 billion reads	Requires HPC or cloud compute to process
Typical patient cohort size	100-500 for rare diseases	Small-N, high-D: your key challenge

DNA Methylation — The Core Feature of Your Classifier

3.1 What Epigenetics Means

The word 'epigenetics' comes from Greek — 'epi' means 'above' or 'upon'. Epigenetics refers to changes in gene activity that do NOT involve changes to the DNA sequence itself. Think of the genome as source code. Epigenetics is the configuration layer — the settings that determine which parts of the code run, in which cells, at which times.

Every cell in your body carries the same DNA sequence. Yet a liver cell looks nothing like a neuron, and neither looks like a blood cell. This is entirely explained by epigenetics — different cells have different epigenetic patterns that silence some genes and activate others, creating distinct cell identities from identical source code.

In cancer, the epigenetic programme breaks down. Genes that should be silenced (oncogenes) become active, and genes that should be active (tumour suppressors) become silenced. This happens through changes in DNA methylation and other epigenetic marks.

3.2 What DNA Methylation Is Physically

DNA methylation is the addition of a methyl group (CH₃) to the 5th carbon position of a cytosine base, producing 5-methylcytosine (written as 5mC or m5C). This is a covalent chemical modification — it physically changes the cytosine molecule, but does not change which base it is. The DNA sequence is preserved; only the methylation state changes.

In humans, DNA methylation occurs almost exclusively at **CpG dinucleotides** — positions where a cytosine is immediately followed by a guanine on the same DNA strand (written 5'-CG-3', hence 'CpG'). The 'p' refers to the phosphodiester bond connecting them.

Physical Reality Check: A methyl group is tiny — just one carbon and three hydrogen atoms. Yet this tiny modification changes how proteins interact with the DNA at that position. When a CpG in a gene's promoter is methylated, proteins that need to bind there to initiate transcription are blocked. The gene effectively gets switched off.

3.3 CpG Sites and CpG Islands

You would expect CpG dinucleotides to appear randomly throughout the genome at the frequency predicted by the base composition. Strangely, CpGs are actually underrepresented across most of the genome — roughly 4-5x fewer than expected. This is because methylated CpGs are mutation hotspots: methylated cytosine spontaneously converts to thymine over evolutionary time, slowly depleting CpGs from methylated regions.

However, certain regions — called **CpG islands** — have maintained a high density of CpGs despite this evolutionary pressure. This is because CpG islands are usually unmethylated, so their CpGs were protected from mutation. CpG islands cluster near gene promoters — approximately 60-70% of human gene promoters contain a CpG island.

Term	Definition	Relevance to Your Work
CpG site	A single position in the genome where cytosine is followed by guanine. ~28 million in human genome.	Each CpG site = one potential feature dimension in your methylation feature vector.
CpG island	A region of >200bp with high CpG density and high G+C content. ~29,000 in human genome.	Promoter CpG islands are the most informative features — highest signal-to-noise.
CpG shore	The 2kb flanking region on either side of a CpG island. Still has elevated CpG density.	Shore methylation is often more variable than islands and highly tissue-specific.
CpG shelf	The 2kb region beyond the shore (2-4kb from island edges).	Lower information content; generally lower priority for feature selection.
Open sea	All remaining CpG sites not near an island, shore, or shelf. Majority of genome.	Often globally hypomethylated in cancer; useful for detecting genome-wide changes.

3.4 How Methylation Controls Gene Expression

Methylation at a gene's promoter CpG island silences that gene through two mechanisms:

Direct blockade: Transcription factors need to bind to specific DNA sequences in the promoter to initiate RNA synthesis. When CpGs in these binding sites are methylated, the transcription factors can no longer recognise and bind the sequence. No binding = no transcription = gene off.

Methyl-binding proteins: Proteins called MBDs (methyl-CpG binding domain proteins) specifically recognise methylated CpGs and recruit chromatin remodelling complexes that compact the DNA, making it physically inaccessible for transcription.

3.5 Cancer-Specific Methylation Patterns

Cancer cells show two types of methylation change compared to normal cells, and both happen simultaneously in AML:

Focal hypermethylation: Specific CpG islands that are normally unmethylated — particularly at tumour suppressor gene promoters — become heavily methylated. This silences genes whose job is to prevent uncontrolled cell division. Example: CDKN2A (p16), a cell cycle brake, is frequently hypermethylated in AML.

Global hypomethylation: The overall level of methylation across the genome decreases in cancer cells. Normally methylated repetitive elements become unmethylated, causing genomic instability. Oncogenes in open sea regions may become activated.

Critically for your ML task: each AML subtype has a characteristic and reproducible pattern of both hypermethylation and hypomethylation. The combination of these patterns across thousands of CpG sites creates a unique molecular fingerprint — and this fingerprint is stable enough across patients with the same subtype to be used as a classifier feature.

3.6 Methylation as a Feature Vector

After running the computational pipeline (covered in Chapter 4), each patient sample is represented as a numerical vector. Each position in the vector corresponds to one CpG site in the genome. The value at each position is the **beta value** — a number between 0 and 1 representing the fraction of DNA molecules at that site that are methylated.

```
# What your data actually looks like per patient: # Each row = one CpG site;  
columns = chromosome, position, beta value chr1 10469 0.92 # heavily methylated  
chr1 10471 0.89 # heavily methylated chr1 10484 0.03 # nearly unmethylated chr1  
10489 0.07 # nearly unmethylated ... # 28 million such rows per patient sample #  
Your feature matrix: patients x CpG_sites (after selection: 500 x 10,000)
```

Beta values cluster bimodally — most sites are either near 0 (unmethylated) or near 1 (methylated). Values in the 0.2-0.8 range represent heterogeneity — either the sample is a mixture of cell types, or there is allele-specific methylation (one chromosome methylated, the other not). This heterogeneity is itself a feature that carries information.

Nanopore Sequencing — Where Your Data Comes From

4.1 The Physical Mechanism

Oxford Nanopore Technology (ONT) is a company that developed a fundamentally different approach to DNA sequencing compared to all earlier technologies. Understanding the physical mechanism is essential because it directly determines what data you receive and what its characteristics are.

The core component is a protein nanopore — a barrel-shaped protein with a central channel approximately 1-2 nanometres in diameter. This pore is embedded in a synthetic lipid membrane. An ionic current is maintained across this membrane by applying a voltage. The current flows through the nanopore channel.

When a DNA molecule is threaded through the pore, the bases physically occupy the channel and disrupt the ionic current. Each combination of bases (a k-mer, typically 5-6 bases) produces a characteristic current level. As the DNA strand moves through the pore at a controlled speed (managed by a motor protein), the current signal traces out a time series that encodes the entire sequence.

Data Reality: The raw output of a nanopore sequencer is NOT a sequence of letters. It is a time-series of electrical current values measured in picoamperes (pA), sampled at ~4,000 times per second. Converting this electrical signal into DNA letters (A, T, G, C) is a machine learning problem solved by basecalling software.

4.2 Why Nanopore Can Detect Methylation Directly

Here is the key insight: when a methylated cytosine (5mC) passes through the nanopore, the electrical current signal at that position is measurably different from an unmethylated cytosine. The methyl group adds mass and changes the charge distribution in a way that shifts the current level by a detectable amount.

This means that a single nanopore sequencing run captures both the DNA sequence (which base is at each position) AND the methylation state (whether the cytosine is methylated) in the same raw signal. No additional chemistry, no extra experiment — the information is already there in the electrical trace.

Contrast with traditional methods: Bisulfite sequencing — the previous gold standard for methylation — requires treating DNA with sodium bisulfite, a harsh chemical that converts unmethylated cytosines to uracil (which reads as thymine). Methylated cytosines are protected from this conversion. By comparing converted vs unconverted positions, you infer methylation. But bisulfite destroys over 90% of the DNA in the process, requires a separate sequencing experiment, and cannot detect modifications other than 5mC.

4.3 Long Reads vs Short Reads

Most widely-used sequencing technologies (Illumina) produce short reads — typically 150 base pairs. The genome must be shredded into millions of tiny fragments, each sequenced independently, and then computationally reassembled like a jigsaw puzzle. Nanopore produces long reads — often 5,000 to

100,000+ base pairs in a single continuous read.

Long reads offer specific advantages for this project:

Resolving repetitive regions: Short reads cannot be uniquely mapped to repetitive sequences (they match multiple locations). Long reads span across repeats and can be placed unambiguously.

Detecting structural variants: Chromosomal translocations (important for AML subtyping) are much easier to detect in long reads that span the breakpoint.

Haplotype phasing: Long reads can determine which mutations are on the same chromosome copy (allele), enabling allele-specific methylation analysis.

Reduced computational assembly complexity: Longer contiguous reads require less assembly work to reconstruct the original genome sequence.

4.4 The Basecalling Step

Basecalling is the process of converting the raw electrical current time-series into a DNA sequence string. This is a well-defined ML problem: the input is a 1D time series (current values over time) and the output is a sequence of characters (DNA bases).

Oxford Nanopore's basecalling software, **Dorado** (the current version, replacing the older Guppy), uses a neural network trained on millions of known sequences to perform this translation. The architecture is based on a Transformer with convolutional layers, trained with Connectionist Temporal Classification (CTC) loss — the same approach used in speech recognition.

Dorado also simultaneously performs methylation calling during basecalling, outputting a probability that each cytosine is methylated based on the same raw signal features. The output is a modified FASTQ file (called a BAM with modification tags) that contains both the sequence and per-base methylation probabilities.

4.5 Data Quality — What to Watch For

Read quality score (Q-score): A measure of basecalling accuracy. Q10 means 90% accuracy per base, Q20 means 99%, Q30 means 99.9%. Modern Dorado achieves Q20+ in high accuracy mode. A minimum Q10 filter is standard.

Coverage depth: How many reads cover each position in the genome. For reliable methylation calling, you need at least 20-30x coverage. Low coverage positions should be excluded from your feature vector or imputed carefully.

Read length N50: The read length at which 50% of total sequenced bases are in reads of that length or longer. A higher N50 indicates better data quality for structural variant detection.

The Computational Pipeline — From Blood to Features

This chapter traces the complete journey of a patient sample from blood draw to the feature matrix that enters your PyTorch model. Knowing this pipeline in full — even steps you won't personally code — allows you to debug problems, understand data quality issues, and have informed conversations with the bioinformatics team.

5.1 Upstream Steps (Not Your Code, But Your Responsibility to Understand)

Step	What Happens	Output	Tool
Blood draw	Patient's blood or bone marrow aspiration taken in clinic.	Biological sample	Clinical procedure
Cell isolation	Leukaemic blast cells are isolated from the blood sample to enrich for cancer cell DNA.	Purified cell population	FACS / density gradient
DNA extraction	Cells are lysed and DNA is purified. For nanopore, high molecular weight (HMW) DNA is critical — fragmented DNA produces short reads.	Purified DNA (HMW preferred)	Kit-based extraction
Library preparation	DNA is prepared for sequencing: repair of damaged ends, addition of sequencing adapters that allow the DNA to thread into the pore.	Sequencing library	ONT Ligation Sequencing Kit
Sequencing run	Library is loaded onto a MinION or PromethION flow cell. Electrical signals recorded in real time.	Raw .pod5 or .fast5 files	MinION / PromethION

5.2 Basecalling and Quality Control

Tool: Dorado (Oxford Nanopore's official basecaller, GPU-accelerated)

Dorado takes the raw .pod5 files and produces a BAM file containing the DNA sequence of each read, per-base quality scores, and per-base methylation probabilities (for 5mC and other modifications). You run Dorado on a GPU — it is computationally intensive.

After basecalling, quality control is performed using **NanoPlot** or **PycoQC** to assess the read length distribution, quality score distribution, and total yield. Reads below a minimum quality threshold (usually Q10) are filtered out.

5.3 Reference Alignment

The sequencer reads the DNA in random fragments — there is no inherent order. To know WHERE each fragment came from in the genome, every read must be aligned to the **reference genome**. The human reference genome is a consensus sequence assembled from many individuals. The current standard is **GRCh38 (hg38)**, released in 2013 with regular patches. It covers ~3.1 billion bases across

24 chromosomes.

Tool: minimap2 — the standard aligner for long reads. It uses a seed-and-extend strategy with minimizer hashing to efficiently align reads millions of bases long. Output is a SAM file (text) or BAM file (binary compressed). You will use **samtools** to sort, index, and inspect BAM files.

```
# Typical alignment command: minimap2 -ax map-ont -y GRCh38.fa
sample_reads.fastq | \ samtools sort -o aligned.bam samtools index aligned.bam #
Check alignment statistics: samtools flagstat aligned.bam # You want >90% of
reads mapping; <85% suggests a problem
```

5.4 Methylation Calling

Tool: Modkit (ONT's official methylation processing tool)

Modkit takes the aligned BAM file (which already contains per-read methylation probabilities from Dorado) and aggregates them per CpG site across all reads. For each CpG position in the genome, it computes the methylation frequency — the fraction of reads that show methylation at that position. The output is a **BEDmethyl** file.

```
# BEDmethyl format (tab-separated): # chrom start end name score strand ...
coverage freq chr1 10469 10470 m 0 + ... 45 0.911 chr1 10471 10472 m 0 + ... 43
0.884 chr1 10484 10485 m 0 - ... 38 0.026 # 'coverage' = number of reads covering
this CpG site # 'freq' = fraction of those reads showing methylation (your beta
value) # Filter: keep only sites with coverage >= 20
```

5.5 Feature Engineering

This is where your ML work begins. You have a BEDmethyl file per patient — up to 28 million rows. You need to convert this into a clean, consistent feature matrix.

Coverage filtering: Remove CpG sites with fewer than 20 reads. Low-coverage sites have unreliable beta values (a site covered by 3 reads where 1 is methylated gives $\text{freq}=0.33$, which is meaningless).

Site selection: Compute the variance of beta values across all patients at each CpG site. Sites with near-zero variance are uninformative — they are the same in every patient and every subtype. Keep the top 5,000-50,000 most variable sites.

Imputation: Some sites will be missing in some patients (filtered out due to low coverage). You must impute these — median imputation per site is a reasonable starting point; more sophisticated methods use KNN or matrix factorisation.

Normalisation: Apply quantile normalisation to make the distribution of beta values comparable across samples. This corrects for systematic differences between sequencing runs.

Batch correction: If samples were sequenced in different batches (different dates, different flow cells), technical batch effects can dominate biological signal. ComBat is the standard tool for correcting this in genomic data.

5.6 The Transcriptome (RNA-seq) Arm

In parallel with methylation, the project also involves **whole genome transcriptome sequencing** — sequencing all RNA in the cell to measure gene expression. RNA is first converted to complementary

DNA (cDNA) and then sequenced.

The pipeline is: FASTQ from RNA-seq -> alignment with **STAR** (a splice-aware aligner, because RNA has introns removed) -> quantification with **featureCounts** or **Salmon** -> a gene expression matrix (patients x genes) -> differential expression analysis with **DESeq2**. This gives you a second feature set complementary to methylation.

The ML Task — Properly Defined

6.1 Formal Problem Definition

Now that you understand the biology and the data, you can define the ML problem precisely:

Given a methylation beta value matrix X of shape $(N \times D)$ where N is patients and D is the number of selected CpG sites, and a corresponding label vector y where each label is an AML subtype (or normal/non-AML), learn a function $f: X \rightarrow y$ that generalises to new patients not seen during training.

The constraints that make this hard: N is small (100-500), D after selection is 5,000-50,000, class balance is uneven (some subtypes are rarer), and the model must be interpretable to clinicians. Deployment requires calibrated probabilities, not just argmax predictions.

6.2 Why You Cannot Just Throw a Deep Network At It

This is a crucial point for a DL engineer to understand and articulate. A ResNet trained on ImageNet has 1.2 million training images. A BERT model is trained on billions of words. You have, at most, a few hundred patients. An unconstrained deep network with millions of parameters will memorise those patients perfectly and completely fail on new ones.

The right approach is to treat this as a constrained optimisation problem. Start simple (logistic regression, random forest), establish a baseline, understand which features matter, then add model complexity only if simpler models plateau and you have evidence that more parameters will help rather than hurt.

6.3 Recommended Modelling Strategy

Stage	Approach	Why
Baseline	Logistic Regression with L2 regularisation; Random Forest	Interpretable; reveals which CpG features are informative; establishes a performance floor
Intermediate	Gradient Boosting (XGBoost / LightGBM); SVM with RBF kernel	Strong tabular data performance; built-in feature importance; handles high-D well
Deep Learning	MLP in PyTorch with BatchNorm, Dropout; early stopping; small architectures (2-3 layers)	Can capture non-linear interactions; needs careful regularisation at small N
Advanced	Fine-tune DNABERT or Nucleotide Transformer; 1D-CNN for genomic context awareness	Pre-trained genomic representations reduce effective sample size requirement
Ensemble	Combine predictions from multiple models (stacking or soft-voting)	Reduces variance; often best final performance at small N

6.4 Evaluation — What Clinical Grade Means

In clinical ML, the cost of different types of errors is asymmetric. Missing a high-risk AML subtype (false negative) is potentially fatal. Misclassifying a subtype (false positive for a different subtype) leads to wrong treatment. Your evaluation must reflect these stakes:

Stratified k-fold cross-validation: With small N , you must use k-fold ($k=5$ or 10). Stratified means each fold preserves the class proportion. Never use a simple train/test split — you waste too much

data.

Per-class sensitivity and specificity: Not just overall accuracy. Report these for every AML subtype. A model that achieves 95% overall accuracy by ignoring the rarest subtypes is clinically useless.

AUC-ROC per class (one-vs-rest): Threshold-independent. Essential for communicating model performance to clinicians who will set their own thresholds.

Calibration: The model's output probability should match the true frequency of that class. If the model says 80% confidence for subtype X, approximately 80% of such predictions should be correct. Use Platt scaling or isotonic regression if calibration is poor.

No data leakage: All preprocessing decisions (feature selection variance thresholds, normalisation parameters, ComBat correction) must be computed ONLY on training data and then applied to test data. Fitting on all data before splitting is one of the most common and serious mistakes in clinical ML.

Glossary — Every Term You Need to Know

This glossary is not a list of words to memorise. Read each entry fully. The goal is that when you hear any of these terms in conversation, you know exactly what the other person is referring to — and can respond with substance.

5-Methylcytosine (5mC)

The chemically modified form of cytosine produced when a methyl group is added to its 5th carbon. This is the most common form of DNA methylation in mammals. The presence of 5mC at a CpG site is what your nanopore-based methylation calling is detecting.

Technical note for ML engineers: When you report methylation frequencies in your BEDmethyl output, the 'm' modification tag specifically refers to 5mC. Dorado also detects 5-hydroxymethylcytosine (5hmC), a related modification with different biological significance, which can be a confounding factor.

Beta value

A number between 0 and 1 representing the methylation level at a specific CpG site in a specific sample. Calculated as: $\text{methylated reads} / (\text{methylated reads} + \text{unmethylated reads})$. A beta value of 0.0 means completely unmethylated; 1.0 means fully methylated.

Technical note for ML engineers: This is the actual number that goes into your feature matrix. Be aware that beta values are bounded (0 to 1), which makes them behave differently from unbounded features in some ML contexts. M-values ($\log_2$ ratio) are statistically better for linear models but less interpretable. Beta values are preferred for reporting.

Basecalling

The computational process of converting raw electrical current signals from the nanopore device into DNA sequence letters (A, T, G, C). Performed by Dorado. This is fundamentally a sequence-to-sequence ML problem — input is a 1D current time-series, output is a character string.

Technical note for ML engineers: The quality of basecalling directly affects the quality of all downstream analysis. Errors in basecalling propagate through alignment, methylation calling, and into your feature vector. Always check Q-score distributions after basecalling.

Bisulfite sequencing

The traditional method for measuring DNA methylation genome-wide. Treats DNA with sodium bisulfite, which converts unmethylated cytosines to uracil (sequenced as thymine) while leaving methylated cytosines unchanged. By comparing to the reference, methylation is inferred.

Technical note for ML engineers: The limitation: bisulfite treatment destroys >90% of the DNA and requires a dedicated sequencing experiment separate from genome sequencing. Nanopore's direct detection obsoletes this method for most purposes. You should know bisulfite sequencing exists because the existing published literature on AML methylation uses it.

BEDmethyl

A file format output by methylation calling tools (Modkit, Megalodon) that reports per-CpG-site methylation frequencies after aggregating across all reads. Built on the BED format standard. Columns include chromosome, position, modification type, coverage depth, and methylation frequency (your beta value).

Technical note for ML engineers: This is your primary input file format for feature engineering. One BEDmethyl file per patient. Reading and parsing these efficiently requires pandas or polars; naive row-by-row iteration is too slow for 28 million rows.

Chromosomal translocation

A structural mutation where a segment of one chromosome breaks off and attaches to a different chromosome. This often creates a fusion gene that produces an abnormal protein driving cancer. Example: t(8;21) in AML joins chromosomes 8 and 21, creating the AML1-ETO fusion gene.

Technical note for ML engineers: Translocations are much easier to detect in long-read nanopore data than short-read data because a single long read can span the breakpoint. Structural variant callers like Sniffles or SVABA detect these from your BAM file. Translocation status can be an additional feature for your classifier alongside methylation.

ComBat

A statistical method for removing batch effects from genomic data using empirical Bayes estimation. It models the batch effect as an additive and multiplicative term and corrects for both. Originally developed for microarray data but standard for methylation data.

Technical note for ML engineers: Use ComBat from the sva R package or the pycombat Python implementation. Apply it after normalisation, before feature selection. Critically: ComBat requires knowing which batch each sample belongs to. If batch and subtype are perfectly confounded (all subtype A samples in batch 1), ComBat cannot separate the effects.

Coverage / Sequencing depth

The average number of sequencing reads that map to each position in the genome. Expressed as 'Nx' — for example, 30x means on average each base is covered by 30 reads. Higher coverage increases confidence in variant calls and methylation frequency estimates.

Technical note for ML engineers: For whole genome methylation analysis, 30x is the standard minimum. For targeted panels of specific genes, 100x or higher is used. In your pipeline, filter out CpG sites with coverage below 10-20 before computing beta values.

Differential methylation

A statistical analysis that identifies CpG sites where methylation levels differ significantly between two or more groups. Methods like t-tests, Wilcoxon tests, or dedicated tools like DSS or methylKit are used. Output is a list of differentially methylated positions (DMPs) or regions (DMRs).

Technical note for ML engineers: Differential methylation analysis is your primary feature selection strategy. Run it pairwise for each AML subtype vs all others. The union of the most significant DMPs across all comparisons forms your selected feature set.

Epigenetics

The study of heritable changes in gene expression that do not involve changes to the DNA sequence itself. The main epigenetic mechanisms are DNA methylation, histone modifications, and non-coding RNA regulation.

Technical note for ML engineers: The word 'epigenetic landscape' refers to the complete state of all epigenetic marks in a cell. Cancer fundamentally reprograms the epigenetic landscape. Your methylation classifier is essentially learning to distinguish different epigenetic landscapes.

GRCh38 / hg38

The current reference assembly of the human genome, released by the Genome Reference Consortium in 2013 (with patches). Contains approximately 3.1 billion base pairs across 24 chromosomes. The standard reference for all current human genomics work.

Technical note for ML engineers: When you align nanopore reads with minimap2, you align against GRCh38. All genomic coordinates in your BEDmethyl file are positions on GRCh38. The previous version hg19/GRCh37 is still used in older studies; be aware of coordinate differences.

Haematopoiesis

The process by which all blood cells are produced from haematopoietic stem cells (HSCs) in the bone marrow. The process involves progressive differentiation through multiple progenitor stages. AML is a disease of arrested differentiation in the myeloid branch.

Technical note for ML engineers: Understanding haematopoiesis helps you understand why AML methylation patterns look the way they do. Different AML subtypes are 'stuck' at different stages of differentiation, each with a characteristic methylation profile reflecting that developmental stage.

Minimal Residual Disease (MRD)

The small number of cancer cells that remain in a patient after treatment, below the threshold detectable by standard morphology or flow cytometry. MRD positivity after treatment predicts relapse. MRD monitoring requires highly sensitive assays.

Technical note for ML engineers: One important application of your methylation classifier is MRD monitoring: tracking the methylation signature of a patient's specific AML subtype in serial blood samples over time to detect early relapse. This is a longitudinal monitoring use case that requires different modelling considerations than initial diagnosis.

Modkit

Oxford Nanopore's official command-line tool for processing methylation data from basecalled BAM files. Takes a BAM file with modification tags (output by Dorado) and produces per-site methylation statistics in BEDmethyl format.

Technical note for ML engineers: Modkit replaced Megalodon as the standard tool. It is faster, handles multiple modification types (5mC, 5hmC), and integrates cleanly with Dorado output. You will run this as part of your pipeline before feature engineering.

Chapter 8

Seven Questions — With Answers You Can Reason Through

These are not scripted answers. Read each question, understand WHY it is being asked and what the interviewer wants to learn about you. Then read the answer as an example of the reasoning process — not a script to memorise. If you understand Chapters 1-7, you should be able to reconstruct these answers yourself.

Question 1

Why is DNA methylation a better feature for AML classification than gene expression alone?

Why they ask this: This tests whether you understand the biology deeply enough to justify your approach, not just implement it.

A strong answer covers:

Gene expression (RNA-seq) measures the output of the gene regulatory system — how much of each protein is being made. It is dynamic, changes rapidly with cell state and environment, and is technically noisy because RNA is unstable. Two samples from the same patient taken hours apart can show meaningful expression differences.

DNA methylation, by contrast, measures the regulatory state itself — the settings that determine which genes can be expressed. Methylation is chemically stable, heritable through cell division, and changes slowly. The methylation pattern of a cancer cell reflects its developmental history and the specific epigenetic programme that drove its malignant transformation.

For cancer subtyping, methylation is more reproducible across patients with the same molecular subtype. This has been demonstrated empirically — methylation classifiers outperform expression classifiers for brain tumour classification (the Capper 2018 Nature paper), and similar results are emerging for leukaemia. The combination of both is the strongest approach.

Question 2

How does nanopore sequencing detect DNA methylation, and why is this an advantage over bisulfite sequencing?

Why they ask this: Tests your understanding of the data generation process — critical for knowing what your data actually represents.

A strong answer covers:

In nanopore sequencing, a DNA strand is threaded through a protein pore and the disruption to ionic current is measured. A methylated cytosine (5mC) carries a methyl group that changes the physical properties of that base as it passes through the pore — the current signal at that position is measurably different from an unmethylated cytosine. Dorado's neural network is trained to recognise these current differences and output a probability of methylation for each cytosine, simultaneously with basecalling.

Bisulfite sequencing requires treating DNA with a harsh chemical that destroys over 90% of the input material. It needs a separate sequencing experiment and cannot distinguish 5mC from 5-hydroxymethylcytosine (5hmC). It also does not provide structural variant information.

The nanopore advantage is that a single sequencing run gives you sequence, methylation, and structural variant information together. For a clinical diagnostic pipeline that needs to be fast and use minimal sample material, this single-assay approach is transformative.

Question 3

Walk me through your feature engineering process from a raw BEDmethyl file to a model-ready input.

Why they ask this: Tests hands-on pipeline knowledge. Shows whether you have thought about data quality, not just model architecture.

A strong answer covers:

Starting with one BEDmethyl file per patient, I would build the feature matrix in these steps:

First, load each file and filter by coverage — remove any CpG site with fewer than 20 reads. Methylation frequencies computed from fewer reads have too much sampling variance.

Second, find the intersection of CpG sites covered across all patients. A site missing in too many patients (say, more than 20%) is removed. For the remaining missing values, I impute using the median beta value across patients at that site.

Third, select informative features. Compute the standard deviation of beta values across all training patients at each site. Sites with near-zero variance are the same in everyone and cannot help a classifier. I would keep the top 10,000-50,000 most variable sites, optionally refined by differential methylation analysis comparing subtypes.

Fourth, normalise — apply quantile normalisation to make beta value distributions comparable across samples from different sequencing runs.

Fifth, correct for batch effects using ComBat if samples were processed in different batches. All these steps must be fit on training data only and applied to test data.

Question 4

Your dataset has only 200 patient samples. How does this change how you approach model design compared to ImageNet-scale problems?

Why they ask this: Tests your ability to adapt your DL knowledge to clinical constraints — the most common failure mode of ML engineers in healthcare.

A strong answer covers:

With $N=200$ and $D=10,000+$ features, I am firmly in the $n \ll p$ regime. The primary risk is not underfitting — it is catastrophic overfitting. A standard ResNet-style network has millions of parameters and will simply memorise 200 patients perfectly while learning nothing generalisable.

My approach would be: start with simpler models that have strong inductive biases. Logistic regression with L2 regularisation can work well at this scale. Random Forest and XGBoost are designed for tabular data and handle high dimensionality gracefully with their tree-based feature selection.

If I use PyTorch, I would use very small networks — two or three layers maximum, with aggressive dropout (0.5+), batch normalisation, and early stopping monitored on a validation fold. The number of parameters must be far smaller than the number of samples.

Transfer learning is a strong option: fine-tune a pre-trained genomic language model like DNABERT on my labelled samples rather than training from scratch. This is analogous to fine-tuning BERT for text

classification in a low-resource language.

I would also use ensembling — combining random forest, XGBoost, and a small MLP via soft voting reduces variance and consistently improves performance at small N.

Question 5

What is a batch effect in genomics, how would you detect it, and how would you correct it?

Why they ask this: Batch effects are the most common cause of genomic ML models failing in practice. This is a test of real-world awareness.

A strong answer covers:

A batch effect is a systematic technical difference between groups of samples that were processed separately. In genomics, this arises when samples are sequenced on different dates, different flow cells, by different technicians, or in different laboratory conditions. The sequencing process introduces small but consistent biases that vary between batches — different coverage depths, different noise characteristics, slight differences in library preparation efficiency.

The danger is that if AML subtype labels happen to correlate with batch membership (which can easily happen in a prospective study where patients of different subtypes arrive at different times), a model will learn to classify batches rather than subtypes. It will appear to perform well on cross-validation but fail completely on new patients.

Detection: Run PCA on the beta value matrix and colour points by batch vs by subtype. If batch separates samples better than subtype in the first two PCs, you have a problem.

Correction: Apply ComBat (empirical Bayes method from the sva package), which models and removes batch-related variance while preserving biological variance. For deep learning, adversarial training with a batch discriminator can make representations batch-invariant. I would also include batch as a covariate in any statistical tests.

Question 6

How would you make your model useful to a clinician who has no ML background?

Why they ask this: Clinical ML deployment requires bridging the gap between model output and clinical decision-making. Tests practical thinking.

A strong answer covers:

There are several layers to this.

First, the output format must be clinically meaningful. Rather than outputting a class index, the model should output a ranked list of AML subtypes with calibrated probabilities. The clinician needs to know not just the top prediction but how confident the model is. A prediction of 95% for subtype X is actionable; a prediction of 45% for X vs 40% for Y should trigger manual review rather than automatic treatment assignment.

Second, the model must be interpretable at the patient level. Using SHAP values, I can show which CpG sites drove the prediction for a specific patient. These CpG coordinates can be annotated to gene names using genomic annotation databases, then cross-referenced with known cancer pathways. The clinician sees: 'Elevated methylation at CDKN2A promoter is the primary driver of this subtype call' — which is biologically meaningful to them.

Third, calibration must be validated prospectively. A model that consistently says 80% confidence and is right 60% of the time will erode clinical trust quickly. I would use Platt scaling to calibrate probabilities and report calibration curves in the validation analysis.

Question 7

What is whole genome transcriptome sequencing and how would you integrate it with methylation data in a multi-modal classifier?

Why they ask this: This covers the second arm of the project. Shows you understand why two data types are better than one.

A strong answer covers:

Whole genome transcriptome sequencing — commonly called RNA-seq — sequences all RNA molecules present in a cell at a given moment, giving a genome-wide snapshot of gene expression. The pipeline is: isolate RNA, convert to cDNA, sequence, align to reference with a splice-aware aligner (STAR), quantify reads per gene (featureCounts or Salmon), producing a counts matrix of patients x genes.

Methylation and expression are complementary in a specific biological sense: methylation measures the epigenetic regulatory state — which genes are set to be active or silenced. Expression measures the actual output — which genes are currently producing RNA. For most tumour suppressor genes in cancer, promoter hypermethylation and low expression go together. But not always — some genes are silenced by other mechanisms, and some methylation changes do not affect expression. Using both gives a fuller picture.

For multi-modal ML integration, I would consider three approaches: early fusion (concatenate methylation and expression feature vectors before the model), late fusion (train separate models on each modality and combine predictions at output), or intermediate fusion (separate encoders for each modality that merge at a bottleneck layer, with cross-modal attention). Given small N, late fusion is safest because it allows each modality model to be trained and validated independently before combining, reducing the risk of one modality dominating.

Day 1 Complete | This is your knowledge foundation. Re-read any chapter that feels unclear before Day 2.

Day 2 will cover: Nanopore Toolchain in Practice | Pipeline Coding | Bioinformatics Tools Deep Dive